EV Battery Course — Module 10: Battery Degradation, Ageing and Second Life

Oct 9, 2026 · @Ashishpreet

For a typical Indian private car driven 15,000 km a year, the 60 kWh pack does only about 35 full cycles a year, so time and temperature (calendar ageing) usually wear it out more than cycling does. This module explains the mechanisms, how to read capacity-fade data honestly, warranty thresholds, health estimation and what happens to a pack after its vehicle life.

1. Learning objectives

- Distinguish calendar ageing from cycle ageing.
- Explain loss of lithium inventory (LLI), loss of active material (LAM), SEI growth, plating, particle cracking and electrolyte decomposition.
- Explain how temperature, SOC, depth of discharge and fast charging affect life.
- Read capacity-vs-cycle graphs and tell lab results from real-world data.
- Separate true degradation from measurement error and changing conditions.
- Explain warranty thresholds, SOH estimation and second-life applications.

2. Fundamentals

- **Capacity fade:** fewer Ah available → less range.
- **Power fade:** higher internal resistance → less power, more heat, slower charging.
- **Calendar ageing:** happens while the battery sits, driven by time, temperature and SOC.
- **Cycle ageing:** happens from use, driven by throughput, depth of discharge, C-rate and temperature.
- **End of life (EOL) for EVs:** conventionally 70–80% of original capacity.

3. Degradation mechanisms

Mechanism	What happens	Main drivers	Effect
SEI growth	Electrolyte keeps reacting at the anode, thickening the SEI	High temperature, high SOC, time	Consumes lithium (LLI); raises resistance
Lithium plating	Metallic lithium deposits on the anode	Fast charging, low temperature, high SOC	LLI, dendrites, safety risk
Particle cracking	Repeated expansion/contraction cracks particles	Deep cycles, high C-rate, nickel-rich and silicon materials	LAM; fresh surfaces grow more SEI
Cathode degradation	Structural change, transition-metal dissolution, oxygen loss	High voltage (high SOC), high temperature	LAM; resistance rise
Electrolyte decomposition	Solvent and salt break down, generating gas	High temperature and voltage	Gas, swelling, resistance rise
Current collector corrosion / binder degradation	Contact loss	Overdischarge, moisture, age	Resistance rise, LAM

LLI vs LAM is the core diagnostic split: LLI means the cell has less cyclable lithium; LAM means less electrode material is available to host it. Both reduce capacity; their signatures on voltage curves differ (see incremental capacity analysis below).

The "knee." Many cells fade slowly and linearly, then suddenly accelerate (a knee point), often when lithium plating or electrolyte depletion takes over. Predicting the knee is a major research and warranty problem.

4. Diagram: illustrative capacity-fade curves

```
xychart-beta
  title "Illustrative capacity vs cycles (lab, 25 °C, 1C/1C, 100% DOD)"
  x-axis "Equivalent full cycles" [0, 500, 1000, 1500, 2000, 2500, 3000]
  y-axis "Capacity retention (%)" 60 --> 100
  line [100, 97, 95, 93, 91, 89, 87]
  line [100, 94, 90, 86, 82, 74, 64]
```

Upper line: an LFP cell (gradual fade). Lower line: an NMC cell with a knee after about 2,000 cycles. These are illustrative shapes, not data from a specific product.

How to read such graphs

1. Check the test conditions: temperature, C-rates, depth of discharge, rest periods. Lab tests at 25 °C and full cycles are not your customer's usage.
2. Check the y-axis start: a graph starting at 90% exaggerates fade; one starting at 0% hides it.
3. Look for the knee and whether tests ran long enough to find it.
4. Count samples: one cell is an anecdote; manufacturers test many cells and report spread.
5. Cycles in lab tests are often accelerated (continuous, no rest); calendar effects are underrepresented.

5. Step-by-step: what ages a pack over a car's life (Indian example)

1. Parked at 100% SOC in 40 °C sun → fastest calendar ageing (SEI growth, cathode stress).
2. Daily commute of 30–40 km → shallow cycles, minimal cycle ageing.
3. Occasional DC fast charges on highway trips → modest plating risk if the pack is cold; heat stress if hot.
4. Thermal management keeps it in range while driving and charging, but not always while parked with the car off.
5. Over 8 years, total throughput is only about 280 full cycles, so calendar ageing dominates.

6. Engineering equations

Calendar fade (empirical square-root-of-time model):

```
Q_{loss}(t) = k\,\sqrt{t}
```

Temperature acceleration (Arrhenius):

```
\frac{k_2}{k_1} = \exp\left[\frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)\right]
```

E_a = activation energy (J/mol, often 30–60 kJ/mol for calendar fade), R = 8.314 J/(mol·K), T in K.

Equivalent full cycles (EFC):

$$EFC = \frac{\text{total energy throughput}}{\text{nominal pack energy}}$$

State of health:

$$SOH_{Q} = \frac{C_{now}}{C_{new}} \qquad SOH_{R} = \frac{R_{now}}{R_{new}}$$

7. Worked examples

Example 1: annual cycles. 15,000 km × 0.145 kWh/km = 2,175 kWh per year ÷ 61.44 kWh = **35 EFC per year**; over 8 years ≈ **283 EFC**. LFP rated for 3,000+ cycles: cycling is not the limit; calendar ageing is.

Example 2: calendar fade. A pack loses 5% in its first year. With Q_loss = k√t: after 4 years: 5 × √4 = **10%**; after 8 years: 5 × √8 = **14.1%**.

Example 3: temperature effect. E_a = 50 kJ/mol.

- 25 → 35 °C: exp[(50,000/8.314) × (1/298.15 − 1/308.15)] = **1.92×** faster.
- 25 → 45 °C: **3.55×** faster.

This is the physics behind the rule of thumb that ageing roughly doubles every 10 °C.

Example 4: SOH. A pack measured at 52.0 kWh under a standard test, new value 61.44 kWh: SOH = 52.0 ÷ 61.44 = **84.6%**.

8. Separating real degradation from noise

Source of apparent "loss"	Why it is not real degradation	How engineers handle it
Cold weather	Lower capacity and higher resistance that recover when warm	Compare at a reference temperature
Different discharge rate	Less capacity at high C-rate	Use a standard C-rate (e.g. C/3)
BMS SOC/SOH estimate error	Estimation drift, not cells	Use measured Ah between known points; reference performance tests
Cell imbalance	Weakest cell limits pack; others have capacity left	Balance, then retest
Rest and relaxation	OCV not settled	Standard rest times

Source of apparent "loss"	Why it is not real degradation	How engineers handle it
Capacity "recovery" after rest	Lithium redistributes into overhang areas	Test protocol control

In the lab, ageing is tracked by periodic **reference performance tests (RPT)**: capacity at fixed temperature and C-rate, pulse resistance, sometimes impedance spectroscopy. Diagnostic methods such as **incremental capacity analysis (dQ/dV)** and **differential voltage analysis (dV/dQ)** reveal whether LLI or LAM is dominant from shifts in curve peaks.

In the field, analysts use fleet data: hundreds or thousands of vehicles, normalised for temperature and usage, with statistical models rather than single-vehicle readings.

9. Warranty and health estimation

- Many EV makers offer about 8 years / 160,000 km battery warranties, typically guaranteeing about 70% of capacity (terms vary by OEM, market and model; check the specific warranty).
- SOH for warranty is usually calculated by the BMS and confirmed by dealer diagnostics or a capacity test.
- Independent SOH certificates are emerging to support used-EV sales; the EU battery regulation will require SOH information in battery passports.

10. Second life

At 70–80% SOH, a pack still holds a lot: 61.44 kWh at 70% = **43 kWh**. Second-life uses:

- Stationary storage: behind-the-meter for commercial buildings, solar smoothing, telecom tower backup, EV charging buffers.
- Lower-demand mobility: e-rickshaws or forklifts.

Challenges: variable SOH across packs, testing cost, different pack designs (CTP/CTB are hard to repurpose), liability and warranty, BMS access and data, and competition from new LFP storage cells now priced around $70/kWh at pack level for storage (BNEF 2025), which squeezes second-life economics.

11. Real-world EV examples

- Early Nissan Leaf (passively cooled) showed faster fade in hot climates; it became the classic lesson in thermal management.

- Large fleet studies (for example, published analyses of telematics data by Geotab and Recurrent in the US) have found average fade far slower than early fears, with liquid-cooled packs often retaining around 85–90% after several years. These are averages, with wide spread by climate and usage.
- Tesla and other OEMs limit default daily charge to about 80–90% for NMC packs, but recommend 100% for LFP packs (for SOC calibration).
- Indian two-wheelers face high heat and frequent full charges; LFP adoption is partly a response.

12. Manufacturing applications

- **Formation and ageing (Module 12)** set initial SEI quality, which influences lifetime fade.
- **Self-discharge screening:** high-self-discharge cells age differently and indicate defects; they are rejected after ageing tests.
- **Cell matching:** reduces imbalance-driven apparent fade.
- **Warranty cost modelling:** use fade models plus fleet usage profiles to estimate the share of packs falling below the warranty threshold; this feeds pricing.

13. Common mistakes and misconceptions

- **"Cycle life rating = years of life."** Calendar life often dominates for private cars.
- **"Never charge above 80%."** True for many NMC packs if parked long at high SOC; LFP is more tolerant and needs periodic full charges.
- **"One cold-weather range figure proves degradation."** Temperature effects are reversible.
- **"Fade is linear forever."** Watch for knees.
- **"Second-life is always economic."** It depends on new-cell prices, testing and repackaging costs.

14. Practical case study: early fade in a ride-hailing fleet

Problem. A ride-hailing fleet in Hyderabad sees SOH of 82% after 18 months, versus 92% expected.

Analysis. Usage: 250 km/day, twice-daily DC fast charging in the afternoon heat, vehicles parked at 100% for hours at the depot. EFC: 250 × 0.145 × 540 days ÷ 61.44 ≈ 319 EFC, plus severe calendar stress (high SOC + heat) and hot fast charging.

Actions. Charge to 90% at depot and finish before departure; schedule fast charging when the pack is cool or precondition it; enable battery cooling while parked; retrain drivers on preconditioning.

Lesson. Usage profile matters as much as chemistry. Fleet contracts should include battery-care rules.

15. Revision notes

- Capacity fade (range) and power fade (resistance) are different measures.
- LLI (lithium lost, mostly SEI and plating) and LAM (host material lost, mostly cracking and dissolution).
- Heat and high SOC drive calendar fade; fast charging and cold drive plating.
- Arrhenius: about 2× faster per +10 °C.
- Compare capacity only under standard conditions (RPT).
- Second life typically starts around 70–80% SOH.

16. Glossary

Term	Meaning
LLI / LAM	Loss of lithium inventory / loss of active material
EFC	Equivalent full cycles
RPT	Reference performance test
ICA / DVA	Incremental capacity / differential voltage analysis
Knee point	Onset of accelerated fade
SOH	State of health
Calendar ageing	Ageing during storage

17. Knowledge check

Conceptual

1. Why does calendar ageing dominate in most private cars?
2. Explain the difference between LLI and LAM.
3. Why does a winter range drop not prove degradation?

4. Why does parking at 100% SOC in heat accelerate ageing?
5. What makes a knee point dangerous for warranty planning?

Numerical

1. A pack loses 3% in the first year ($\sqrt{t}$ model). Predict loss after 9 years.
2. With E_a = 40 kJ/mol, how much faster is ageing at 40 °C than 25 °C?
3. A taxi drives 60,000 km/year at 0.13 kWh/km with a 40 kWh pack. EFC per year?

Application

1. Design a test plan to determine whether field complaints are true degradation or BMS estimation error.
2. Evaluate a business case for using 70%-SOH packs as backup storage at telecom towers.

18. Answer key

Conceptual

1. Annual throughput is small (about 30–40 EFC), while time, temperature and SOC act every day.
2. LLI: cyclable lithium is consumed (SEI, plating), so less lithium shuttles. LAM: electrode host material is lost or disconnected (cracking, dissolution), so less can be stored even with lithium available.
3. Cold reduces capacity and raises resistance reversibly; capacity returns when warm.
4. High SOC means a highly delithiated cathode and a lithium-full anode at high potentials, which speeds electrolyte reactions; heat speeds all reactions (Arrhenius).
5. Fade suddenly accelerates, so packs fall below the warranty threshold much earlier than linear extrapolation suggests.

Numerical

1. $3 \times \sqrt{9}$ = **9%**.
2. $\exp[(40{,}000/8.314) \times (1/298.15 - 1/313.15)] = \exp(4{,}811 \times 0.0001607) = \exp(0.773) \approx$ **2.17×**.
3. 60,000 × 0.13 = 7,800 kWh ÷ 40 = **195 EFC/year**.

Application

1. Retrieve BMS logs (SOC, SOH, temperatures, cell voltages); run a full capacity test at 25 °C and C/3 after balancing; compare measured capacity with BMS SOH; check cell spread; check current-sensor calibration. If measured capacity is fine, it's an estimation problem; if not, investigate usage and thermal history.

2. Compare per-kWh cost of tested, repackaged second-life packs (testing, repackaging, BMS, warranty, logistics) against new LFP storage (about $70/kWh for storage packs in 2025 per BNEF, before system costs); check tower duty (shallow cycles suit second life), temperature environment, safety certification and expected remaining life.

19. Further learning

- Birkl et al., "Degradation diagnostics for lithium ion cells," *J. Power Sources* 2017.
- Attia et al., "Knees in lithium-ion battery aging trajectories," *J. Electrochem. Soc.* 2022.
- Severson et al., "Data-driven prediction of battery cycle life before capacity degradation," *Nature Energy* 2019.
- Recurrent and Geotab public reports on EV battery health (US fleets).

Next: Module 11, testing, validation and quality engineering.